Symmetry in the structure module and the loss

What the Supplementary Information of Jumper et al. (2021) proves, and what it does not claim

Equations are from the SI, with section and equation numbers in grey, pages are printed pages. Drawings are illustrations. Headings in blue belong to Evita's slides 10 and 11, in red to the FAPE slide 12. Black is shared ground.

0. Notation and the one definition that matters (SI 1.1, 1.8, 1.8.2)

A frame is a Euclidean transform $T = (R, \vec{\mathbf{t}})$. It maps local to global coordinates and composes as

$$T \circ \vec{\mathbf{x}} = R\vec{\mathbf{x}} + \vec{\mathbf{t}}, \qquad (R_1, \vec{\mathbf{t}}_1) \circ (R_2, \vec{\mathbf{t}}_2) = (R_1 R_2,\ R_1\vec{\mathbf{t}}_2 + \vec{\mathbf{t}}_1). \qquad \text{(SI 1.1)}$$

The SI never names a group. It defines the word it uses instead: "rigid motion" includes proper rotation and translation but not reflection (SI 1.8.2, p. 28). In group language a rigid motion is an element of SE(3), $\det R = +1$. The larger group E(3) also contains reflections, $\det R = -1$. Every claim below is about SE(3). Reflections are treated separately, because the paper wants them broken: proteins are chiral.

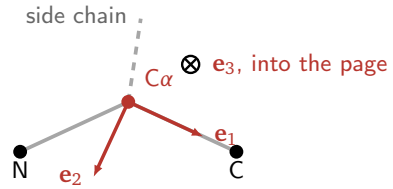
1. Frames are built right-handed (SI 1.8.1, Algorithm 21)

A frame is made from three atoms by Gram–Schmidt. For the backbone, $\vec{\mathbf{x}}_1 = \mathrm{N}$, $\vec{\mathbf{x}}_2 = \mathrm{C}\alpha$, $\vec{\mathbf{x}}_3 = \mathrm{C}$, so $\mathrm{C}\alpha$ is the origin:

$$\mathbf{e}_1 = \frac{\vec{\mathbf{x}}_3 - \vec{\mathbf{x}}_2}{\|\vec{\mathbf{x}}_3 - \vec{\mathbf{x}}_2\|}, \qquad \mathbf{e}_2 = \frac{\mathbf{u}_2}{\|\mathbf{u}_2\|}, \quad \mathbf{u}_2 = (\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2) - \mathbf{e}_1\big(\mathbf{e}_1^\top(\vec{\mathbf{x}}_1 - \vec{\mathbf{x}}_2)\big),$$

$$\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2, \qquad R = (\mathbf{e}_1\, \mathbf{e}_2\, \mathbf{e}_3), \qquad \vec{\mathbf{t}} = \vec{\mathbf{x}}_2.$$

The cross product makes $\det R = +1$ always. This is the one line where chirality enters: reflect all atoms, and the frame does not reflect with them (section 4).



The backbone frame of one residue. $\mathbf{e}_1$ points to C, $\mathbf{e}_2$ is the N direction made orthogonal, $\mathbf{e}_3$ follows by the right-hand rule.

2. IPA is invariant under rigid motions of the frames (SI 1.8.2, Algorithm 22, eqs. 3–6)

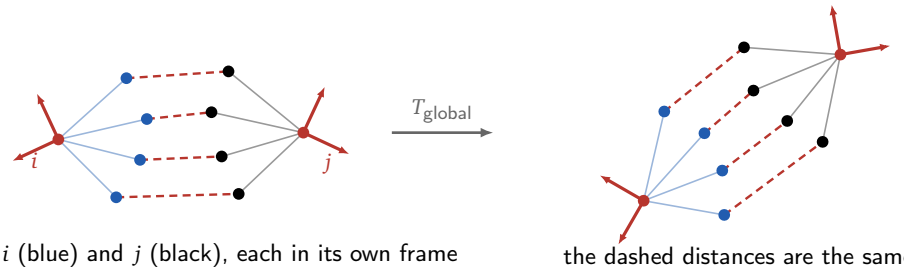
Each residue i has a frame T_i and emits, per head h, query points $\vec{\mathbf{q}}_i^{hp}$ and key points $\vec{\mathbf{k}}_i^{hp}$ in its own frame. The attention logit (Algorithm 22, line 7) is

$$a_{ij}^h = \mathrm{softmax}_j\Big(w_L\Big(\tfrac{1}{\sqrt{c}}\,\mathbf{q}_i^{h\top}\mathbf{k}_j^h + b_{ij}^h - \tfrac{\gamma^h w_C}{2}\sum_p \big\|T_i \circ \vec{\mathbf{q}}_i^{hp} - T_j \circ \vec{\mathbf{k}}_j^{hp}\big\|^2\Big)\Big).$$

Apply one rigid motion T_{global} to every frame. It cancels in the only geometric term, because the norm is invariant under rigid transforms, and again when the output points (line 10) are mapped back into frame i:

$$\big\|(T_{\text{global}} \circ T_i) \circ \vec{\mathbf{q}}_i^{hp} - (T_{\text{global}} \circ T_j) \circ \vec{\mathbf{k}}_j^{hp}\big\|^2 = \big\|T_i \circ \vec{\mathbf{q}}_i^{hp} - T_j \circ \vec{\mathbf{k}}_j^{hp}\big\|^2, \qquad \text{(eqs. 3, 4)}$$

$$(T_{\text{global}} \circ T_i)^{-1} \circ \sum_j a_{ij}^h\big((T_{\text{global}} \circ T_j) \circ \vec{\mathbf{v}}_j^{hp}\big) = T_i^{-1} \circ \sum_j a_{ij}^h\big(T_j \circ \vec{\mathbf{v}}_j^{hp}\big). \qquad \text{(eqs. 5, 6)}$$



points of i (blue) and j (black), each in its own frame — the dashed distances are the same

The only geometry in IPA is the set of distances between points placed in the frames. A rigid motion of all frames moves the points together and leaves the distances unchanged, so the attention weights and the output are unchanged. Exact, by construction, not learned.

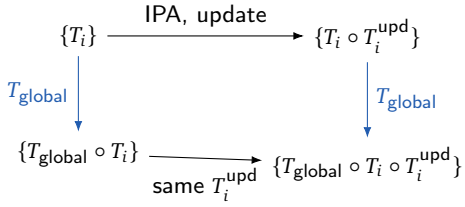
A remark the SI does not make: the two cancellations use only $\|T_{\text{global}} \circ \vec{\mathbf{x}}\| = \|\vec{\mathbf{x}}\|$, which also holds for reflections. IPA as a function of frames is invariant under all of E(3). Chirality is not in IPA. It is in how frames are obtained from atoms (section 1) and in the loss (section 5).

3. The backbone update makes each block equivariant (SI 1.8, 1.8.3, Algorithm 23)

From $\tilde{\mathbf{s}}_i$ alone the network predicts a quaternion $(1, b_i, c_i, d_i)$, normalised, and a translation, giving a small transform T_i^{upd}. The frame is updated in its own local coordinates,

$$T_i \leftarrow T_i \circ T_i^{\text{upd}}. \qquad \text{(Algorithm 20, line 10)}$$

If all frames are moved by T_{global}, section 2 says $\tilde{\mathbf{s}}_i$ and hence T_i^{upd} are unchanged, and $(T_{\text{global}} \circ T_i) \circ T_i^{\text{upd}} = T_{\text{global}} \circ (T_i \circ T_i^{\text{upd}})$ by associativity. The output moves with the input: "this update rule is equivariant under rigid motions" (SI p. 28). Eight such blocks with shared weights compose to an equivariant structure module.



Equivariance is this square commuting: moving the frames first or last gives the same result, because the update is computed in local coordinates.

Two facts that bound the claim. First, all frames start as the identity, the "black hole initialization" (SI 1.8, Algorithm 20 line 4), so there is no input geometry at the start. Equivariance says the module never prefers a global frame. The global frame of the output is arbitrary, which is why the loss has to be invariant (section 5). Second, the Evoformer has no geometry at all. Recycling feeds the previous structure back only as binned Cβ distances (SI 1.10, Algorithm 32), which are invariant under all of E(3), reflection included.

4. A reflection of the atoms is visible in local coordinates (SI 1.9.3, eqs. 10–15)

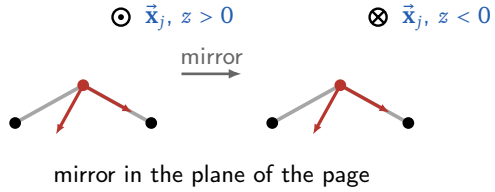
Reflect every atom, $\vec{\mathbf{x}} \mapsto -\vec{\mathbf{x}}$. Rebuilding the frames with Algorithm 21 from the reflected atoms gives

$$T_i = (R_i, \vec{\mathbf{t}}_i) \;\mapsto\; \left(R_i \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, -\vec{\mathbf{t}}_i\right), \qquad \text{(eq. 11)}$$

because $\mathbf{e}_1$ and $\mathbf{e}_2$ flip sign with the atoms while $\mathbf{e}_3 = \mathbf{e}_1 \times \mathbf{e}_2$ does not. The local coordinates of any atom then change by

$$T_i^{-1} \circ \vec{\mathbf{x}}_j \;\mapsto\; \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} (T_i^{-1} \circ \vec{\mathbf{x}}_j). \qquad \text{(eqs. 12–15)}$$

A reflection flips exactly the z component of every local coordinate. Everything that works in local coordinates, IPA and FAPE, can therefore tell a protein from its mirror image.



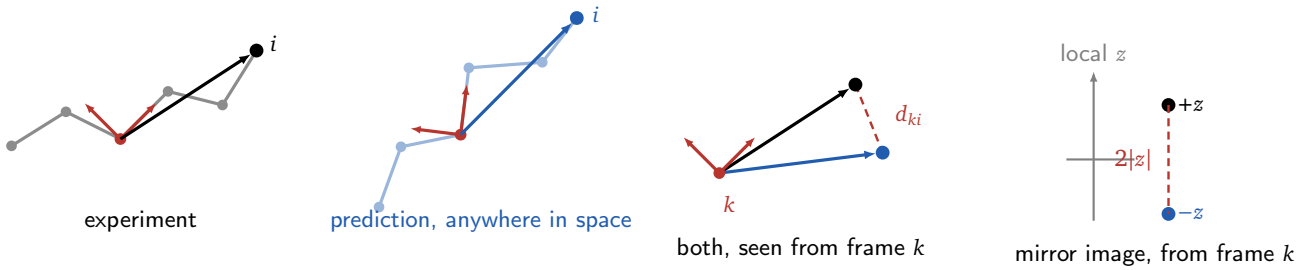
Mirror the protein in the page. The three frame atoms lie in the page and stay, so the frame stays. An atom above the page ($\odot$) moves below it ($\otimes$): its local z changes sign. Any mirror is this one up to a rotation, which the frames absorb.

5. FAPE is invariant under rigid motions and sees chirality (SI 1.9.2, Algorithm 28, eqs. 16–17)

For predicted frames and atoms $\{T_i\}, \{\vec{\mathbf{x}}_j\}$ and the true $\{T_i^{\text{true}}\}, \{\vec{\mathbf{x}}_j^{\text{true}}\}$,

$$\vec{\mathbf{x}}_{ij} = T_i^{-1} \circ \vec{\mathbf{x}}_j, \qquad \vec{\mathbf{x}}_{ij}^{\text{true}} = T_i^{\text{true}\,-1} \circ \vec{\mathbf{x}}_j^{\text{true}}, \qquad d_{ij} = \sqrt{\|\vec{\mathbf{x}}_{ij} - \vec{\mathbf{x}}_{ij}^{\text{true}}\|^2 + \epsilon}, \qquad \mathcal{L}_{\text{FAPE}} = \tfrac{1}{Z} \operatorname{mean}_{i,j} \left(\min(d_{\text{clamp}}, d_{ij})\right),$$

with $Z = d_{\text{clamp}} = 10\,\text{Å}$. Every atom is scored in every frame, $N_{\text{frames}} \times N_{\text{atoms}}$ terms.



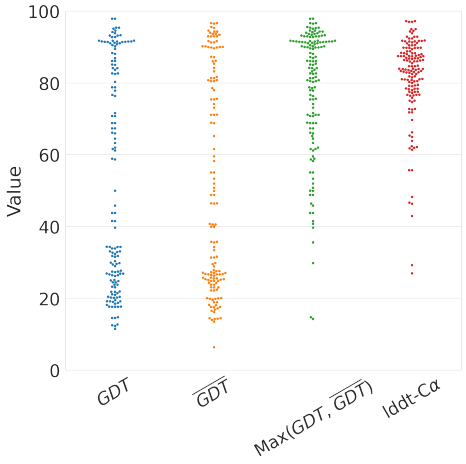
Left and middle: the prediction may sit anywhere, FAPE compares it with the experiment in the local frame k, so a rigid motion of the whole prediction changes nothing. Right: the mirror image has the same shape but, by section 4, every local z flips, so each atom is off by $2|z|$ and the loss is large.

Invariance: a rigid motion applied to the prediction moves T_i and $\vec{\mathbf{x}}_j$ together, and $\vec{\mathbf{x}}_{ij} = T_i^{-1} \circ \vec{\mathbf{x}}_j$ is unchanged, by the same cancellation as in section 2. The same holds for the ground truth. So the loss "will stay constant" under any rotation and translation of either structure (SI p. 34). Not invariance: by section 4,

$$\text{FAPE}(X, X_{\text{reflection}}) = \sum_{ij} \left\| T_i^{-1} \circ \vec{\mathbf{x}}_j - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} (T_i^{-1} \circ \vec{\mathbf{x}}_j) \right\| = 2 \sum_{ij} \left| (T_i^{-1} \circ \vec{\mathbf{x}}_j)_z \right|, \qquad \text{(eqs. 16, 17)}$$

"a large positive value outside of degenerate cases". The ground truth frames come from the experimental atoms with the right-handed Algorithm 21, so the network is trained towards the correct hand.

Two further results. (SI 1.9.4, eqs. 18–22): with S the RMSD after the best alignment over SE(3), $\text{FAPE}(X, \tilde{X}) \geq S(\{\vec{\mathbf{x}}_j\}, \{\tilde{\vec{\mathbf{x}}}_j\})$ regardless of the frames, so FAPE is zero only if the two point sets are related by a proper rigid motion, and a chiral pair always has positive loss, however the frames are built. (SI 1.9.5, eqs. 23–30): idealised FAPE is a pseudometric, non-negative, symmetric, with the triangle inequality. Evidence that this is what fixes the hand (SI 1.9.3, Suppl. Fig. 9): a model trained with dRMSD, a loss that is reflection invariant, reaches good lDDT-Cα, which cannot see chirality, but its GDT is bimodal, and the low mode is the mirror image: $\overline{\text{GDT}}$, the GDT against the reflected target, is high exactly there. "FAPE is the main component that ensures the correct chirality of predicted structures." Side chains add a second source: atoms are built from the frames and the predicted χ angles with ideal geometry, which "will always generate a left-handed molecule" (SI p. 35), so the network cannot build the mirror image of a side chain at all.



Suppl. Fig. 9: a model trained with dRMSD instead of FAPE, 87 CASP14 domains. GDT splits into two modes. The low mode scores high against the mirrored target ($\overline{\text{GDT}}$). lDDT-Cα cannot see the difference.

6. Summary

Component	Under rigid motions, SE(3)	Under reflection	Source
Evoformer, recycling	no geometry, distances invariant	invariant	SI 1.10
Frames from atoms	move with the atoms	do not reflect, eq. 11	SI 1.8.1, 1.9.3
IPA	invariant, exact	invariant, given the frames	SI 1.8.2
Backbone update, one block	equivariant, exact	as the frames	SI 1.8.3
FAPE	invariant	not invariant, eq. 17	SI 1.9.2, 1.9.3
All-atom construction	equivariant	always the correct hand	SI 1.8.4, p. 35

What to say if asked "is it SE(3)-equivariant": yes, the structure module is exactly equivariant under rotations and translations and the loss is invariant, both by working in local frames. It is deliberately not equivariant under reflections, so the output has a definite handedness, learned from the data. What not to say: that AlphaFold uses equivariant kernels or spherical harmonics as in SE(3)-Transformers. The mechanism is only that points are expressed in per-residue frames, so a global motion cancels. The SI proves this in four lines, equations 3 to 6.